

CODE:

```
//Application of Cramer's rule
#include<iostream>
#include<cmath>
#include<iomanip>
using namespace std;

double det(int n, double a[100][100])
{
    int i, j, k;
    //pivotalisation code
    double determinant=1.0;
    for(i=0; i<n; i++)
        for(k=i+1; k<n; k++)
            if(abs(a[i][i])<abs(a[k][i]))
            {
                for(j=0; j<n; j++)
                {
                    double temp=a[i][j];
                    a[i][j]=a[k][j];
                    a[k][j]=temp;
                }
                determinant*=-1;
            }
    //Diagonalization code
    for(i=0; i<n; i++)
        for(j=i+1; j<n; j++)
        {
            double t=a[j][i]/a[i][i];
            for(k=0; k<n; k++)
                a[j][k]=a[j][k]-t*a[i][k];
        }
    //Finally, calculate the determinant
    for(i=0; i<n; i++)
        determinant*=a[i][i];
    return determinant;
}

int main()
{
    int n, i, j, k;
    cout.precision(2);
    cout.setf(ios::fixed);
    cout<<"\nEnter the order of the matrix\n";
    cin>>n;
    double CF[100][100], b[100], CM[100][100], x[n], T[100][100];
    cout<<"\nEnter the elements of the matrix row-wise\n";
```

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        for(i=0;i<n;i++)
            for(j=0;j<n;j++)
                cin>>CF[i][j];
cout<<"\nEnter the elements of the RHM\n";
        for(i=0;i<n;i++)
            cin>>b[i];
//Store the co-efficient matrix in a different matrix
        for(i=0;i<n;i++)
            for(j=0;j<n;j++)
                T[i][j]=CF[i][j];
        if(det(n,CF)==0)
        {
            cout<<"\nThe system doesn't a unique solution\n";
            exit(0);
        }
        else
        {
            //Now apply the cramer's rule
            for(i=0;i<n;i++)
            {
                for(j=0;j<n;j++)
                {
                    for(k=0;k<n;k++)
                    {
                        if(k!=i)
                            CM[j][k]=T[j][k];
                        else
                            CM[j][k]=b[j];
                    }
                }
                x[i]=det(n,CM)/det(n,CF);
            }
        }
        cout<<"\nThe required solution is\n";
        for(i=0;i<n;i++)
            cout<<x[i]<<endl;
        return 0;
    }

```

OUTPUT:

Enter the order of the matrix

3

Enter the elements of the matrix row-wise

1

2

3
2
-1
1
3
1
-2

Enter the elements of the RHM

6
2
2

The required solution is

1.00
1.00
1.00

Process returned 0 (0x0) execution time : 36.621 s
Press any key to continue.

CODE:

```
//Use of Gauss Elimination to solve system of linear equation
#include<iostream>
#include<iomanip>
#include<cmath>
using namespace std;
int main()
{
    int n,i,j,k;
    cout.precision(4);    //set precision
    cout.setf(ios::fixed);
    cout<<"\nEnter the no. of equations\n";
    cin>>n;    //input the no. of equations
    float a[n][n+1],x[n];    //declare an array to store the elements of augmented-matrix
    cout<<"\nEnter the elements of the augmented-matrix row-wise:\n";
    for (i=0;i<n;i++)
        for (j=0;j<=n;j++)
            cin>>a[i][j];    //input the elements of array
    for (i=0;i<n;i++)    //Pivotisation to make the equations diagonally dominant
        for (k=i+1;k<n;k++)
            if (abs(a[i][i])<abs(a[k][i]))
                for (j=0;j<=n;j++)
                {
                    double temp=a[i][j];
                    a[i][j]=a[k][j];
                    a[k][j]=temp;
                }
    cout<<"\nThe matrix after Pivotisation is:\n";
    for (i=0;i<n;i++)    //print the new matrix
    {
        for (j=0;j<=n;j++)
            cout<<a[i][j]<<setw(16);
        cout<<"\n";
    }
    for (i=0;i<n-1;i++)    //loop to perform the gauss elimination
        for (k=i+1;k<n;k++)
        {
            double t=a[k][i]/a[i][i];
            for (j=0;j<=n;j++)
                a[k][j]=a[k][j]-t*a[i][j];    //make the elements below the pivot elements equal to
            //zero or eliminate the variables
        }

    cout<<"\n\nThe matrix after gauss-elimination is as follows:\n";
    for (i=0;i<n;i++)    //print the new matrix
    {
        for (j=0;j<=n;j++)
```

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        cout<<a[i][j]<<setw(16);
    cout<<"\n";
}
for (i=n-1;i>=0;i--)          //back-substitution
{
    if(a[i][i]==0)
    {
        cout<<"\nThe system is inconsistent\n";
        exit(0);
    }
    else
    {
        x[i]=a[i][n];
        for (j=i+1;j<n;j++)
        if (j!=i)
            x[i]=x[i]-a[i][j]*x[j];
        x[i]=x[i]/a[i][i];
    }

}

cout<<"\nThe values of the variables are as follows:\n";
for (i=0;i<n;i++)
    cout<<x[i]<<endl;
return 0;
}

```

OUTPUT:

Enter the no. of equations

3

Enter the elements of the augmented-matrix row-wise:

2

1

-1

8

-3

-1

2

-11

-2

1

2

-3

The matrix after Pivotisation is:

-3.0000	-1.0000	2.0000	-11.0000
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2.0000	1.0000	-1.0000	8.0000
--------	--------	---------	--------

-2.0000	1.0000	2.0000	-3.0000
---------	--------	--------	---------

The matrix after gauss-elimination is as follows:

-3.0000	-1.0000	2.0000	-11.0000
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-0.0000	0.3333	0.3333	0.6667
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0.0000	0.0000	-1.0000	1.0000
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The values of the variables are as follows:

2.0000

3.0000

-1.0000